Yan-Han Chen

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ACADEMIC POSITIONS

Postdoctoral Research Associate — Jul 2026 – Present
Aerospace Engineering, Iowa State University — Ames, IA
Advisor: Prof. Abhishek Halder
Focus: Sample-based learning method for Schrödinger bridge problem, and its application in machine learning

EDUCATION

Ph.D., Statistics — 2020 – 2026
Iowa State University — Ames, IA
Advisors: Prof. Arka P. Ghosh, Prof. Hailiang Liu, and Prof. Ruoyu Wu
Focus: Optimal Transport and Interacting Particle Systems
Thesis: Topics in Stochastic Processes

M.S., Statistics — 2014 – 2016
National Chiao Tung University — Hsinchu, Taiwan
Advisor: Prof. Hui-Nien Hung

B.S., Applied Mathematics — 2010 – 2014
National Chiao Tung University — Hsinchu, Taiwan

RESEARCH INTERESTS

Schrödinger bridge, Interacting Particle Systems, Optimal Transport and their intersection with machine learning

PUBLICATIONS AND PREPRINTS

5. [In Preparation] **Chen Y.-H.**, Ghosh A., Wu R. "Limit theorems for heterogeneous graphon JSQ(d) model." Near completion; draft available upon request.
4. [Journal] **Chen Y.-H.**, Liu H. (2026). "Neural solver for Wasserstein Geodesics and optimal transport dynamics." *Journal of Scientific Computing*, 108(2), 39.
3. [Journal] Thorn S., Chao A., Georgiev K. B., Müller J., Bässler C., Campbell J. L., Castro J., **Chen Y.-H.**, Choi C.-Y., Cobb T. P., et al. (2020). "Estimating retention benchmarks for salvage logging to protect biodiversity." *Nature Communications*, 11(1), 4762.
2. [Journal] Thorn S., Chao A., Bernhardt-Römermann M., **Chen Y.-H.**, Georgiev K. B., Heibl C., Müller J., Schäfer H., Bässler C. (2020). "Rare species, functional groups, and evolutionary lineages drive successional trajectories in disturbed forests." *Ecology*, 101(3), e02949.
1. [Journal] Georgiev K. B., Chao A., Castro J., **Chen Y.-H.**, Choi C.-Y., Fontaine J. B., Hutto R. L., Lee E.-J., Müller J., Rost J., et al. (2020). "Salvage logging changes the taxonomic, phylogenetic and functional successional trajectories of forest bird communities." *Journal of Applied Ecology*, 57(6), 1103–1112.

HONORS AND AWARDS

Teaching Excellence Award May 2026
Iowa State University Ames, IA

Travel Award May 2026
ASA Midwest Regional Conference in Statistics and Data Science

Travel Award Oct 2025
SIAM Central States Section Annual Meeting

Government Fellowship for Studying Abroad 2020 – 2023
Ministry of Education, Taiwan
Full fellowship covering tuition and living expenses for the first three years of Ph.D. study (up to US$150,000).

Funded Participant, Summer School on Mathematical Finance Jul 2014
Institute of Mathematics, Academia Sinica Taipei, Taiwan

Funded Participant, Summer School on Probability Jul 2013
Mathematics Division, National Center for Theoretical Sciences Hsinchu, Taiwan

ADDITIONAL RESEARCH EXPERIENCE

Corporate Research Assistant Aug 2022 – Aug 2025
Corteva Agriscience
Supervisor: Dr. Carl Walker
Projects:

- Adaptive experimental design on fertilizer data using Bayesian optimization with Gaussian process regression
- Reinforcement learning for controlling water level in a two-tank system.

Research Assistant Aug 2017 – Jun 2020
National Tsing Hua University Hsinchu, Taiwan
Supervisor: Prof. Anne Chao
Research:

- Nonparametric methodology in statistical ecology, including diversity estimation, diversity decomposition and dissimilarity, and extensions to hierarchical structures
- Analysis of ecological data with proposed methodology
- Improvement on the efficiency of existing packages for proposed methodology.

TEACHING EXPERIENCE

Instructor
Department of Statistics, Iowa State University

– STAT 3300: Probability and Statistics for Computer Science Summer 2026
– STAT 5280: Visual Business Analytics Fall 2025
– STAT 226: Introduction to Business Statistics I Spring 2022

Lab Assistant / Grader
Department of Statistics, Iowa State University

– STAT 330: Probability and Statistics for Computer Science Summer 2022
– STAT 326: Introduction to Business Statistics II Spring 2022
– STAT 305: Engineering Statistics Summer 2021

PRESENTATIONS

Neural Solver for Wasserstein Geodesics and Optimal Transport Dynamics

ASA Midwest Regional Conference in Statistics and Data Science, Minneapolis, MN — *Poster* May 2026

SIAM Central States Section Annual Meeting, Fayetteville, AR — *Contributed talk* Oct 2025

SIAM Great Lakes Section Meeting, Chicago, IL — *Mini-session talk* Sep 2025

Limit Theorems for Heterogeneous Graphon JSQ(d) Model

INFORMS Annual Meeting, Seattle, WA — *Invited talk* Oct 2024

Analysis and Probability Seminar, Iowa State University — *Seminar talk* Oct 2024

SERVICE AND OUTREACH

Analysis and Probability Reading Group Fall 2024 – Spring 2026
Iowa State University

– Weekly meetings on weak convergence of probability measures, large deviations, and interacting particle systems.

Volunteer Tutor, CyMath Jan 2024 – Dec 2024
Iowa State University

– Weekly math tutoring for students in grades 3–12.

Substitute Military Service (Education Service) Aug 2016 – Jul 2017
Taiwan

– Tutored and mentored underprivileged junior high school students in mathematics.